An explicit estimate for Garcia's paired floor sum

Principia Math In Progress

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Task 1 (original problem). MathDB 357173 records Conjecture 6.6 of Garcia [1, p. 30]. For positive integers n with $p = 4n - 1$ prime, put

$$M = \left\lfloor \frac{n^2}{p} \right\rfloor, \qquad R_m = \sqrt{mp}, \qquad Q_m = \frac{1 + \sqrt{4mp + 3p - 4}}{2},$$

and

$$A_n = \sum_{m=1}^{M} \lfloor R_m \rfloor + \sum_{m=0}^{M-1} \lfloor Q_m \rfloor, \qquad D_n = Mp + 2n.$$

Prove that $A_n/D_n \to 1/3$. The source also proposes $\lfloor D_n/3 \rfloor$ as a *good estimate* without specifying an error. The following theorem proves the limit and gives a precise relative-error interpretation of that estimate.

Theorem 1. *For every integer $n \geq 4$, with the definitions above and without any primality assumption,*

$$\left| A_n - \frac{D_n}{3} \right| \leq 4n, \qquad |A_n - \lfloor D_n/3 \rfloor| \leq 4n + 1, \qquad \left| \frac{A_n}{D_n} - \frac{1}{3} \right| \leq \frac{8}{n}.$$

In particular the conjectured limit holds along the prime subsequence.

Proof. First,

$$\frac{n^2}{4n-1} = \frac{n}{4} + \frac{n}{4(4n-1)}, \qquad 0 < \frac{n}{4(4n-1)} < \frac{1}{4}.$$

The fractional part of $n/4$ is one of $0, 1/4, 1/2, 3/4$, so $M = \lfloor n/4 \rfloor$. In particular $M \geq 1$.

Write

$$B = \sum_{m=1}^{M} \lfloor \sqrt{mp} \rfloor, \qquad T = \sqrt{p} \sum_{m=1}^{M} \sqrt{m}, \qquad V = \frac{2}{3}\sqrt{p}\, M^{3/2}.$$

Monotonicity of $\sqrt{x}$ gives

$$V = \sqrt{p} \int_0^M \sqrt{x}\, dx \;\; \leq T \leq V + \sqrt{pM}.$$

For the upper bound, apply the integral comparison to the terms with $m < M$, and then add $\sqrt{M}$. Rounding each summand gives

$$T - M \leq B \leq T.$$

Since $p \geq 15$, for every $0 \leq m < M$,

$$R_m \leq Q_m = \frac{1}{2} + \sqrt{mp + \frac{3p}{4} - 1} \leq R_{m+1} + 1.$$

Consequently, if $C = \sum_{m=0}^{M-1} \lfloor Q_m \rfloor$, then

$$B - \lfloor R_M \rfloor \le C \le B + M.$$

Combining these estimates and writing $W = 2V$ proves the useful sandwich

$$W - 2M - \sqrt{pM} \le A_n \le W + 2\sqrt{pM} + M. \tag{1}$$

This argument retains every endpoint, including integral square roots.

For completeness we turn the sandwich into explicit bounds. Write $r = n - 4M \in \{0, 1, 2, 3\}$. Then

$$W = \frac{n^2}{3}\sqrt{1 - \frac{1}{4n}}\left(1 - \frac{r}{n}\right)^{3/2}.$$

For $0 \le x, y \le 1$,

$$1 - \sqrt{1 - x} \le x, \qquad 1 - (1 - y)^{3/2} \le \frac{3}{2}y.$$

The second inequality follows by integrating the derivative $\frac{3}{2}\sqrt{1 - y} \le \frac{3}{2}$. Since $1 - ab \le (1 - a) + (1 - b)$ for $a, b \in [0, 1]$, we get

$$0 \le \frac{n^2}{3} - W \le \frac{n}{12} + \frac{rn}{2} \le \frac{19n}{12}.$$

Also $M \le n/4$ and $\sqrt{pM} \le n$. Thus (1) yields

$$-\frac{37n}{12} \le A_n - \frac{n^2}{3} \le \frac{9n}{4}, \qquad \left|A_n - \frac{n^2}{3}\right| \le \frac{37n}{12}.$$

Direct expansion gives

$$D_n = n^2 + \frac{(7 - 4r)n + r}{4}, \qquad \left|\frac{D_n}{3} - \frac{n^2}{3}\right| \le \frac{2n}{3}.$$

The last inequality holds for each of the four possible r. It follows that

$$\left|A_n - \frac{D_n}{3}\right| \le \frac{45n}{12} \le 4n.$$

Rounding $D_n/3$ changes it by less than one, proving the second claim. Finally,

$$D_n \ge n^2 - \frac{5n}{4} \ge \frac{n^2}{2} \qquad (n \ge 4).$$

Dividing the first bound by D_n proves the third claim and the limit. □

Scope and verification. The proof uses only the definitions of the two roots, an integral comparison, and bounded rounding errors. It does not assume any of the source's class-number or jump-count identities. The source's qualitative phrase *good estimate* is made precise here as an $O(n)$ absolute error on a quantity of order n^2; no bounded absolute error was specified or is asserted. The accompanying integer-arithmetic verifier checks the original two sums, their root inequalities, and the explicit bounds for every $4 \le n \le 4000$, together with larger examples. These finite checks support the proof but do not establish the infinite limit.

Task 2 (proposed follow-up). Determine the optimal constant in a uniform bound $|A_n - D_n/3| \leq cn + O(1)$.

Task 3 (proposed follow-up). For each residue class of n modulo four, determine whether $(A_n - D_n/3)/n$ converges, and compare unrestricted and prime subsequences.

Task 4 (proposed follow-up). Obtain smaller remainder terms after subtracting any second-order main term, including the effect of square divisors of $4n - 1$.

Task 5 (proposed follow-up). Determine the corresponding limit and explicit error when $p/n \to c > 0$, $M = \lfloor n^2/p \rfloor$, and Q_m is replaced by $\alpha + \sqrt{mp + \beta p + \gamma}$, for fixed real parameters whenever all summands are defined.

References

[1] J. Garcia, *Class Number of the Imaginary Quadratic Field and Quadratic Residues Identities*, arXiv:2301.02951v1 (2023), 31 pages, `https://arxiv.org/abs/2301.02951`. Definitions: Notation 2.2; target: Conjecture 6.6.